Basics of Minimax Algorithm

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Definition

The Minimax algorithm is a decision-making algorithm that is used in two-player games, where the goal is to find the best possible move for the current player while assuming that the other player will also make optimal moves.

2

Working Principle

The algorithm works by creating a game tree that represents all possible moves by both players. It then analyzes the tree, minimizing the maximum possible loss for the player, and maximizing the minimum possible gain.

3

Pseudocode

We can implement the Minimax algorithm using pseudocode. The algorithm recursively traverses the game tree and assigns scores to each move, ultimately returning the best possible move.

Understanding Alpha-Beta Pruning



The Concept

Alpha-Beta pruning is a technique used to improve the efficiency of the Minimax algorithm. It works by pruning branches of the game tree that don't need to be explored further, thereby reducing the total number of nodes that need to be evaluated.



How it Works

The algorithm maintains two values, alpha and beta, that are used to keep track of the best known score of the maximizing and minimizing player, respectively. It then prunes branches that can't affect the final score and limits the search space accordingly.



Performance Boost

Alpha-Beta pruning can significantly reduce the number of nodes evaluated, making the algorithm much faster than the regular Minimax algorithm for large game trees.

Benefits and Limitations

Advantages

- Can be applied to any two-player game
- Efficient in searching game trees
- Maximizes the player’s benefit while minimizing the opponent’s gain

Disadvantages

- Not suitable for games with imperfect information or unknown state space
- Requires a lot of memory to store and traverse the game tree
- Can be computationally expensive in certain cases

Real-World Applications

Gaming

It is widely used in games, from Chess and Go to computer games like Starcraft and World of Warcraft.

Decision Making

These algorithms can be used in decision-making scenarios such as resource allocation and project planning.

Optimization

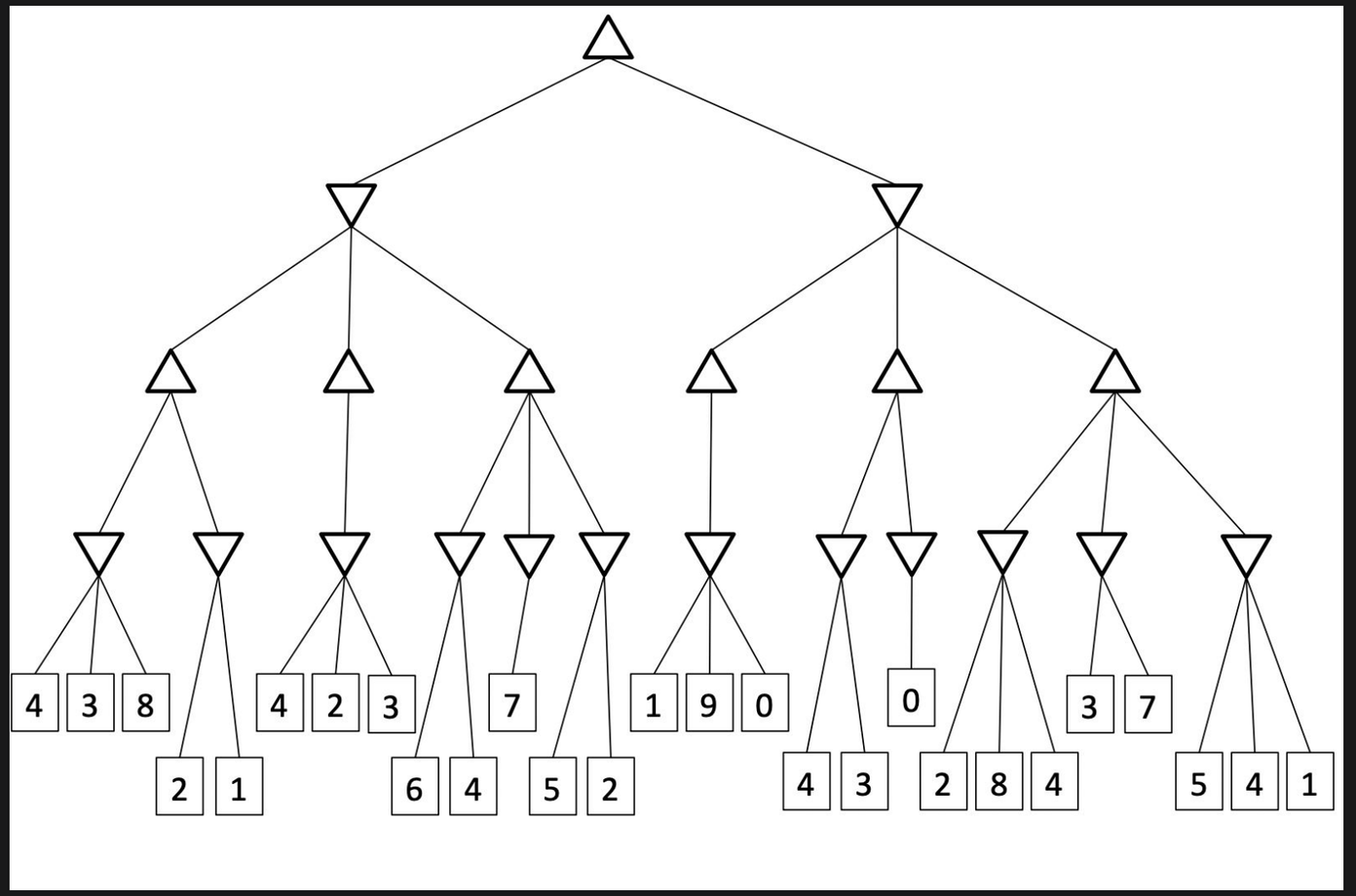
They can be used to optimize complex systems such as traffic flow, energy management, and financial portfolios.

Materials

https://www.youtube.com/watch?v=l-hh51ncgDI

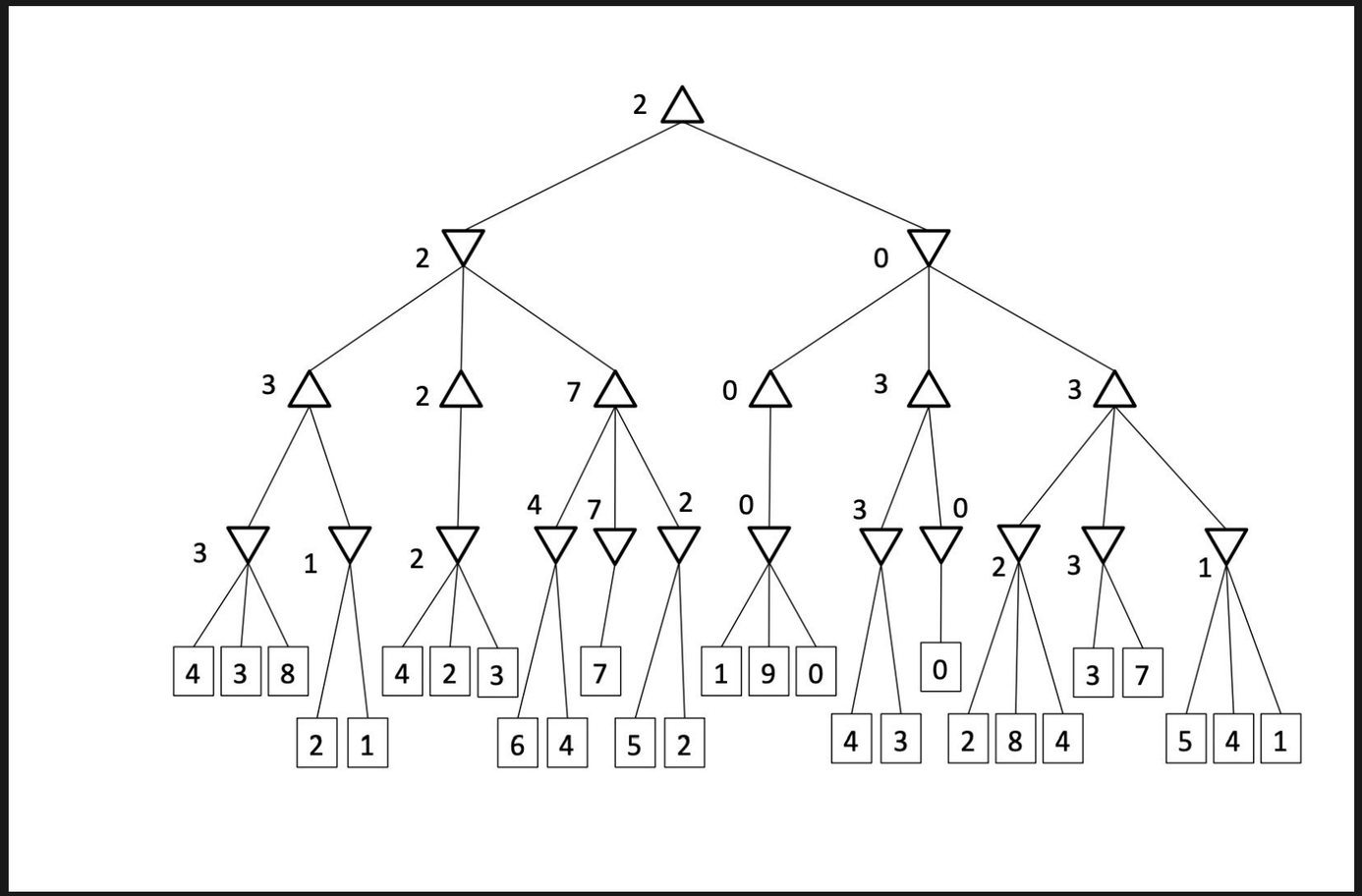
Exercise

Give the values calculated by minimax for all states in the tree.



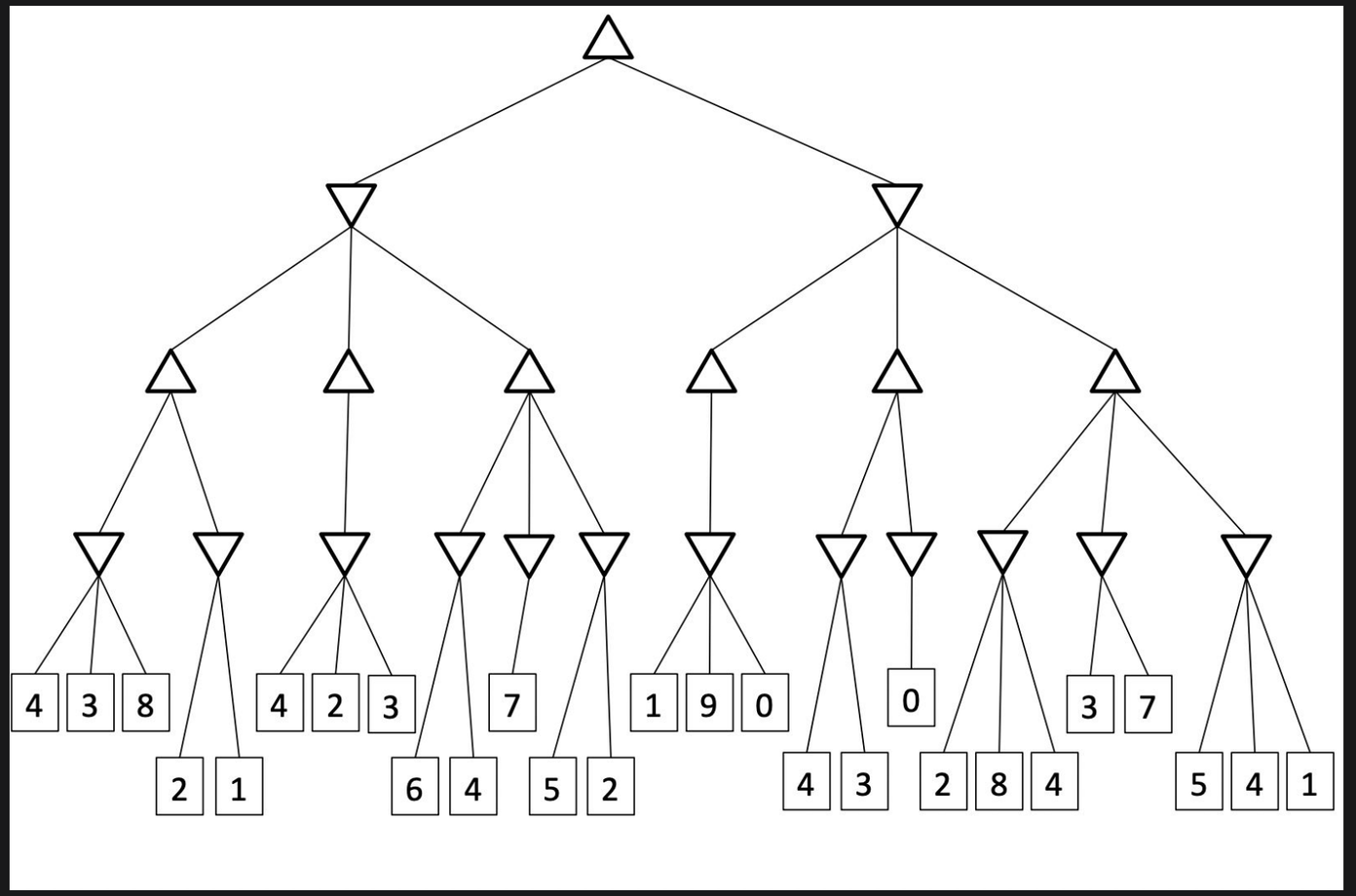
Solution

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Extension

Indicate which branches of the tree will be pruned by alpha-beta pruning.



Solution

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